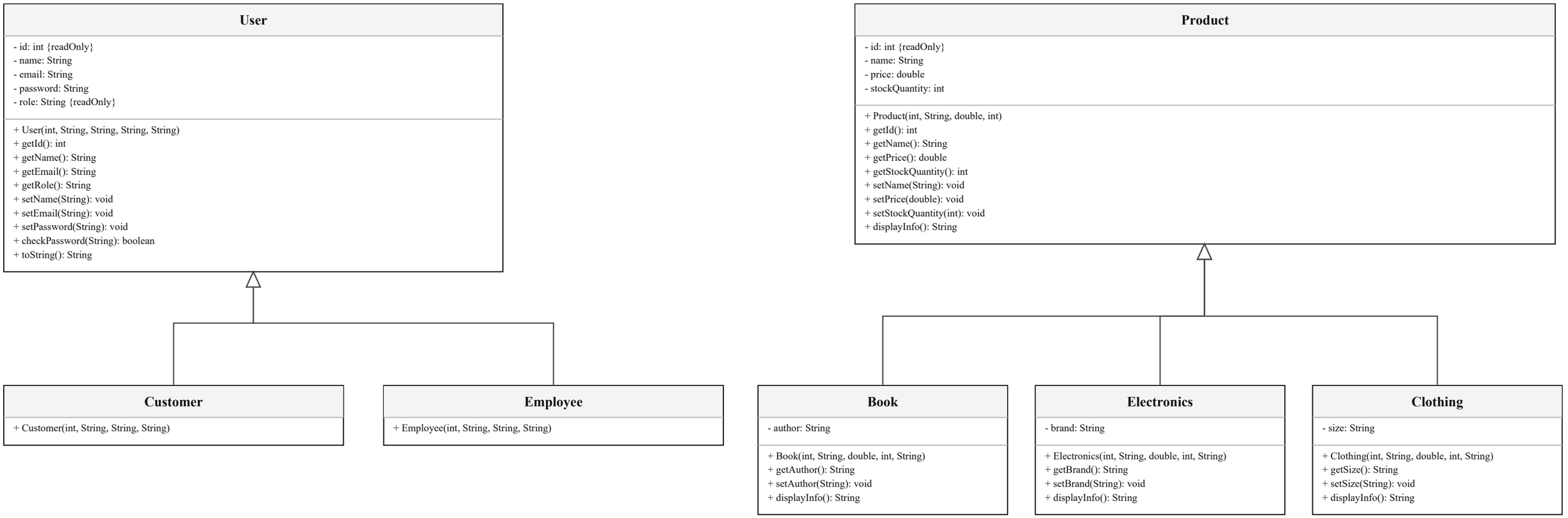
Figure 1. Users and products

Open triangles point from a subclass to its parent.



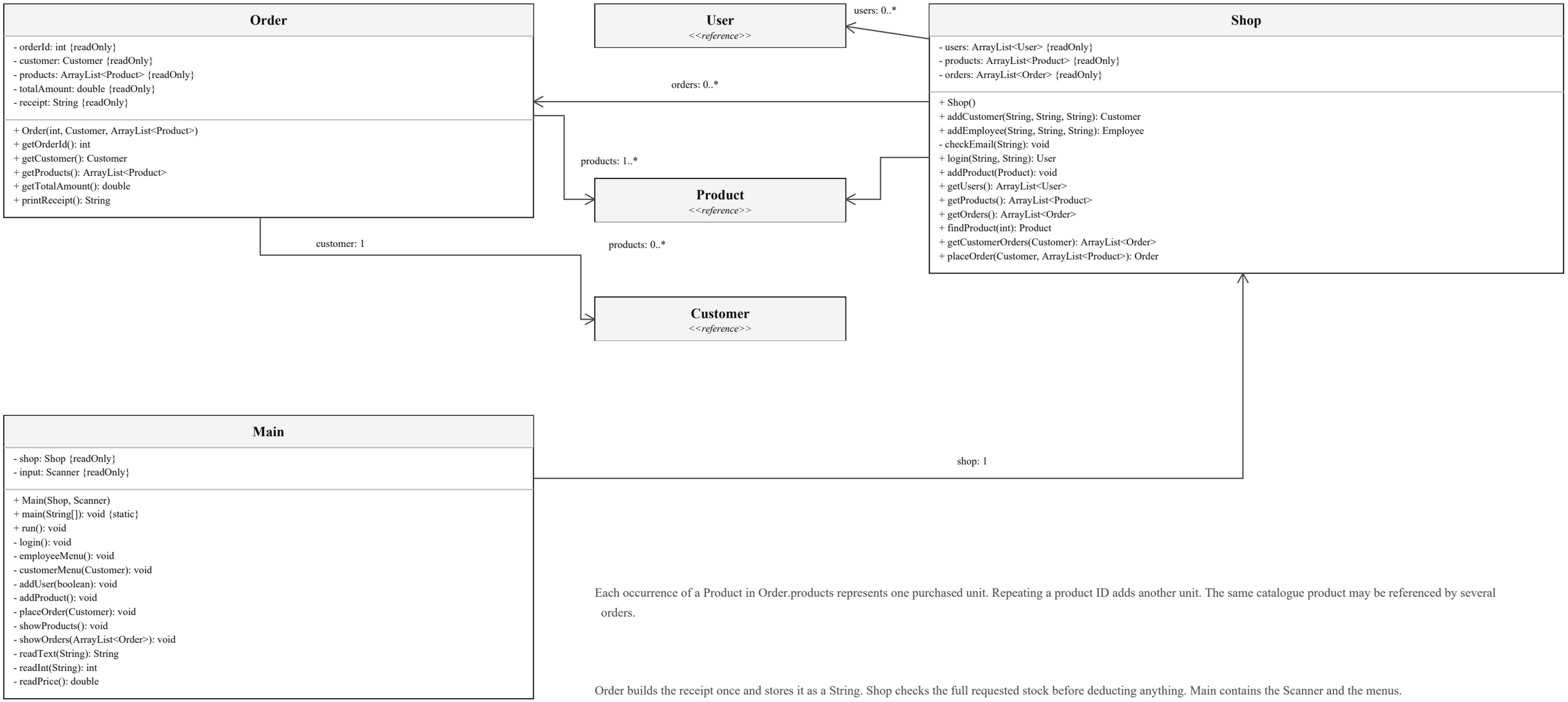
User and Product are ordinary parent classes. Customer and Employee pass their role to the User constructor. Each product subclass adds one field and overrides displayInfo().

Notation: + public, - private. {readOnly} means a final field, and {static} means a class member. Constructor parameters are shown by type. All declared state fields are private.

The model package contains the user, product and order classes. Shop coordinates operations, and Main handles console input.

Figure 2. Orders, shop and main program

Solid arrows show references stored in fields.



Each occurrence of a Product in Order.products represents one purchased unit. Repeating a product ID adds another unit. The same catalogue product may be referenced by several orders.

Order builds the receipt once and stores it as a String. Shop checks the full requested stock before deducting anything. Main contains the Scanner and the menus.

Scanner, ArrayList, String and other Java library types appear as field and parameter types. The optional test runner is separate from the application.